Simplified 157-bus system data

1. Bus

Bus	Type	Load (MW)	Bus	Type	Load (MW)	Bus	Type	Load (MW)
1	1	-259.99	31	1	-90.01	61	2	20.90
2	1	-60.01	32	2	331.56	62	2	291.95
3	1	-129.99	33	2	768.07	63	1	59.92
4	1	-824.25	34	1	328.00	64	1	-149.94
5	1	239.40	35	1	145.33	65	1	199.72
6	2	755.22	36	1	-0.05	66	2	778.99
7	2	72.65	37	1	-0.21	67	1	198.08
8	2	72.61	38	1	0.00	68	2	470.51
9	1	-660.53	39	2	142.19	69	1	161.82
10	1	-497.65	40	4	700.00	70	1	0.00
11	1	-90.00	41	1	-698.50	71	1	2042.97
12	1	0.05	42	1	71.40	72	1	200.78
13	1	-209.81	43	2	138.33	73	2	1.03
14	2	-144.62	44	1	-249.74	74	1	346.08
15	1	-299.68	45	1	-399.13	75	1	-359.70
16	3	72.25	46	2	55.97	76	1	253.88
17	1	0.00	47	1	212.45	77	1	339.50
18	1	0.00	48	1	-346.43	78	2	202.67
19	1	0.00	49	1	47.22	79	2	273.21
20	1	435.76	50	2	58.82	80	2	36.66
21	1	0.00	51	1	0.00	81	1	139.71
22	1	0.00	52	1	0.00	82	2	37.06
23	2	37.06	53	1	719.81	83	1	254.96
24	1	34.02	54	2	167.59	84	2	254.67
25	1	1.92	55	1	378.97	85	2	36.01
26	1	-364.52	56	1	246.73	86	1	-2.20
27	2	778.98	57	1	-0.13	87	4	0.00
28	1	235.87	58	1	0.00	88	1	130.55
29	4	0.00	59	1	0.00	89	1	110.09
30	1	-99.89	60	1	0.00	90	2	37.09

(Type 1=PQ, 2=PV, 3=Ref, 4=isolated; wind farm buses are highlighted by red)

Bus	Type	Load (MW)	Bus	Type	Load (MW)	Bus	Type	Load (MW)
91	1	0.00	121	1	1603.60	151	1	0.00
92	1	236.67	122	4	0.00	152	1	0.00
93	2	0.89	123	4	0.00	153	1	0.00
94	1	253.47	124	4	0.00	154	1	0.00
95	1	369.02	125	4	0.00	155	1	0.00
96	2	579.92	126	2	0.22	156	1	0.00
97	1	281.60	127	1	136.04	157	1	0.00
98	1	44.17	128	1	132.70			
99	1	638.62	129	4	0.00			
100	2	49.57	130	4	0.00			
101	2	106.11	131	4	0.00			
102	2	184.06	132	2	0.40			
103	4	0.00	133	1	0.00			
104	1	113.98	134	1	0.00			
105	1	264.17	135	1	0.00			
106	1	310.94	136	1	0.00			
107	2	237.69	137	1	0.00			
108	1	-255.43	138	1	0.00			
109	1	0.00	139	1	0.00			
110	1	0.00	140	1	58.28			
111	2	792.24	141	1	0.00			
112	2	443.82	142	1	0.00			
113	1	-49.11	143	1	0.00			
114	2	-480.08	144	1	0.00			
115	2	911.97	145	1	0.00			
116	2	0.54	146	1	0.00			
117	4	0.00	147	1	0.00			
118	2	173.22	148	1	0.00			
119	1	-0.02	149	1	0.00			
120	1	1552.33	150	1	0.00			

(Type 1=PQ, 2=PV, 3=Ref, 4=isolated; wind farm buses are highlighted by red)

2. Branch

From	To	X	From	To	X	From	To	X
1	12	0.009	47	35	0.018	79	80	0.001
2	138	0.011	47	83	0.052	79	97	0.007
3	14	0.015	47	136	0.024	79	97	0.007
7	19	0.001	51	32	0.007	81	98	0.025
8	19	0.001	51	32	0.007	81	136	0.017
11	139	0.007	51	50	0.008	81	136	0.016
12	13	0.005	51	50	0.008	82	136	0.005
12	38	0.014	51	52	0.010	82	136	0.005
12	38	0.014	51	52	0.010	83	74	0.055
12	38	0.014	51	63	0.033	83	88	0.011
14	15	0.003	51	63	0.033	83	88	0.011
17	4	0.007	53	52	0.009	83	89	0.011
17	16	0.003	53	52	0.009	83	136	0.015
17	18	0.006	55	25	0.027	83	100	0.018
17	110	0.006	55	54	0.019	84	92	0.003
18	70	0.007	56	43	0.010	84	92	0.003
18	6	0.006	56	43	0.011	86	113	0.013
18	19	0.011	56	66	0.008	89	101	0.018
18	19	0.011	56	66	0.007	91	77	0.003
23	33	0.002	62	61	0.005	91	77	0.003
23	33	0.002	62	72	0.020	91	90	0.006
27	28	0.014	62	72	0.019	91	90	0.006
31	38	0.010	62	63	0.021	91	92	0.006
32	14	0.044	157	61	0.005	91	92	0.006
32	14	0.038	63	62	0.021	91	104	0.002
33	32	0.002	64	51	0.023	91	104	0.002
33	32	0.002	64	51	0.023	93	92	0.000
34	134	0.010	65	54	0.012	93	92	0.000
34	134	0.010	65	66	0.006	96	107	0.001
35	28	0.022	65	78	0.011	96	107	0.001
36	9	0.012	65	78	0.011	98	102	0.013
36	9	0.012	67	66	0.015	99	81	0.001
36	37	0.007	67	66	0.015	99	81	0.002
37	10	0.016	67	68	0.010	101	118	0.018
37	10	0.016	67	68	0.010	101	118	0.018
37	58	0.012	68	134	0.008	104	105	0.019
37	58	0.012	68	69	0.003	104	105	0.019
37	36	0.007	68	134	0.009	106	105	0.018
38	21	0.011	69	134	0.006	106	105	0.019
38	22	0.010	70	18	0.007	106	137	0.020

From	To	X	From	To	X	From	To	X
38	30	0.011	70	71	0.009	106	137	0.022
38	39	0.001	70	71	0.009	109	58	0.017
38	39	0.001	72	73	0.024	109	58	0.017
38	44	0.003	72	76	0.019	110	17	0.006
38	45	0.010	72	76	0.019	110	121	0.007
38	49	0.015	74	136	0.011	110	109	0.008
38	49	0.015	75	91	0.021	110	109	0.008
38	49	0.015	75	91	0.021	111	112	0.006
41	53	0.002	76	91	0.005	112	96	0.016
41	53	0.002	76	91	0.005	112	111	0.006
42	55	0.022	77	84	0.006	112	113	0.019
42	133	0.022	77	84	0.006	113	96	0.007
43	66	0.017	78	85	0.005	113	97	0.002
43	26	0.048	78	85	0.005	113	97	0.002
46	47	0.004	78	86	0.006	113	126	0.009
46	47	0.004	78	96	0.007	114	97	0.001
46	47	0.004	78	112	0.011	114	113	0.001
46	34	0.061	79	68	0.008	114	113	0.001
46	34	0.061	79	68	0.008	114	115	0.023
47	27	0.022	79	80	0.001	114	115	0.024

(Transmission capacity is set to be 150% of the base case branch flow)

3. Generator

Bus	PGmax(p.u.)	PGmin(p.u.)	Bus	PGmax(p.u.)	PGmin(p.u.)
6	8.75	4.81	114	10.44	3.13
7	16.50	9.08	115	6.94	2.08
8	15.00	8.25	116	1.25	0.38
14	1.39	0.42	100	0.46	0.14
16	16.50	9.08	111	3.75	2.06
23	7.50	4.13	112	0.73	0.22
27	4.25	2.34	126	0.60	0.18
32	1.88	1.03	118	1.25	0.38
32	0.45	0.14	132	1.00	0.30
33	5.28	2.90	14	1.38	0.76
39	0.69	0.21	14	1.02	0.31
40	8.75	4.81	32	0.25	0.08
43	20.50	11.28	43	2.75	1.51
46	15.00	8.25	50	4.13	2.27
50	4.13	2.27	54	1.31	0.39
54	0.44	0.13	62	2.18	0.65
61	7.50	4.13	66	3.75	1.13

Bus	PGmax(p.u.)	PGmin(p.u.)	Bus	PGmax(p.u.)	PGmin(p.u.)
62	1.80	0.54	68	2.16	0.65
66	0.94	0.52	73	3.05	0.92
68	2.75	1.51	78	0.60	0.18
68	0.72	0.22	79	1.85	0.56
73	1.70	0.51	96	4.98	1.49
78	0.60	0.18	100	0.93	0.28
79	0.90	0.27	101	0.76	0.23
80	8.25	4.54	102	4.28	1.28
84	3.75	2.06	111	3.75	2.06
85	8.25	4.54	112	0.64	0.19
82	8.25	4.54	115	1.78	0.53
90	8.25	4.54	116	2.50	0.75
93	8.75	4.81	118	3.42	1.03
96	3.55	1.95	126	2.40	0.72
96	1.60	0.48	128	2.40	0.72
101	1.09	0.33	132	2.00	0.60
102	3.10	0.93	140	8.25	4.54
107	11.25	6.19			

(Ramping rate is set to be (PGmax-PGmin)/1.5 (p.u./hr.))